dragons or whatever else was happening then). But, as we discussed before, general relativity and particle physics don't like to work together in their current form, and so we've made even less progress toward a TOE than a GUT. Many people are placing their bets on string theory being the ultimate TOE. But if you thought GUTs were hard to verify experimentally, TOEs may actually be impossible to test, at least with anything like the technology we can currently conceive of. Arguments rage from time to time about whether or not this is the case, and whether or not untestable theories should even be called science. I don't think the situation is quite as dire as that. Cosmology may offer solutions (and no, I'm not just saying that because I'm a cosmologist). In certain cases, with a bit of creativity, there are some tantalizing possibilities for testing predictions of string theory and related ideas with observations of the cosmos. If we make it through a couple of apocalypses in the next few chapters, we'll see how cosmology might be able to show us more about the ultimate, tying-it-all-up-with-a-bow fundamental structure of the universe than any particle experiment can.

But let's get back to our story. We have just escaped the throes of Planck Time–quantum-gravity confusion and are enjoying the fundamental-force unity of the very slightly less speculative GUT era.

COSMIC INFLATION

What happened next is still a matter of debate, but the near-consensus in cosmology is that sometime around this moment the universe suddenly experienced the mother of all growth spurts—a process we call *cosmic inflation*. For reasons we're still trying to understand, the expansion of the universe suddenly went into very high gear, with the region that would someday become our entire observable universe increasing in size by a factor of more than 100 trillion trillion (i.e., 10^{26}). Of course, that only brought it up to about the size of a beach ball, but given that the starting point was unimaginable tininess smaller than any known particle, and the growth happened over the course of something like 10^{-34} seconds, we have reason to be impressed.

The theory of inflation came about to solve a few really perplexing problems in the standard Big Bang model. One has to do with the weird uniformity of the cosmic microwave background, another with its tiny imperfections.